CURRICULUM VITAE

Jiye Kim

lool1015@snu.ac.kr

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APPOINTMENT

2026.09 - Present

Post-doctoral Fellow
Department of Electrical and Computer Engineering
Seoul National University, Seoul, Korea

EDUCATIONS

2020.09 - 2026.08

Ph. D. in Electrical and Computer Engineering
Department of Electrical and Computer Engineering
Seoul National University, Seoul, Korea

Adviser: Professor Jongho Lee

2025.11 - 2026.04

Visiting Student
Athinoula A. Martinos Center for Biomedical Imaging,
Massachusetts General Hospital and Harvard Medical School
Charlestown, USA

Adviser: Professor Berkin Bilgic

2016 – 2020.08

B.S. in Electrical and Computer Engineering
Department of Electrical and Computer Engineering
Seoul National University, Seoul, Korea
Cum Laude Honors

2019

Study Abroad Exchange Program, Electrical Engineering
ETH Zurich, Zurich, Switzerland

HONORS AND AWARDS

- Distinguished Dissertation Award, Seoul National University, 2026
- ISMRM Magna Cum Laude Merit Award, "Simultaneous reconstruction of longitudinal QSM data (Longitudinal-QSM)", 2026
- 2nd Place Abstract Award, ISMRM EMTP Study Group, "Simultaneous reconstruction of longitudinal QSM data (Longitudinal-QSM)", 2026

- Best Oral Presentation 1st, ISMRM EMTP Study Group Trainee Day, “Simultaneous reconstruction of longitudinal QSM data (Longitudinal-QSM)”, 2026

- ISMRM Magna Cum Laude Merit Award, “Visualization of in-vivo brainstem myeloarchitecture via χ-separation at 7T”, 2025

- Best Trainee Scientific Awards Poster Presentation Grand Prix, International Congress on MRI & Annual Scientific Meeting of KSMRM, 2023

- Best Trainee Scientific Awards Poster Presentation Gold, International Congress on MRI & Annual Scientific Meeting of KSMRM, 2022

- Best Trainee Scientific Awards, Oral Presentation 2nd, International Congress on MRI & Annual Scientific Meeting of KSMRM, 2021

- The Encouragement Prize, The Joint Conference of the IBEC and the ICBHI, 2021

- Academic Excellence Scholarship, 2018-2019

- Global Exchange Scholarship, Mirae Asset Park Hyeon Joo Foundation, 2019

Journal Publications

- X. Yong, Z. Liu, Y. Dong, H. Hong, Y. Chen, P. Dubovan, Q. Liu, **J. Kim**, U. D. Gallastegi, Y. Jun, and B. Bilgic, “SeqEyes: An Open-Source Toolbox for Synchronized Waveform/Trajectory Visualization in Pulseq.”, *Magnetic Resonance in Medicine* (2026)

- T. Kim, S. Ji, K. Min, M. Kim, J. Youn, C. Oh, **J. Kim**, and J. Lee, “Vessel segmentation for χ-separation”, *Magnetic Resonance in Medicine* (2025)

- Y. Chen, Y. Jun, A. Heydari, X. Yong, **J. Kim**, J. Lee, ... & Bilgic, B., “MIMOSA: Multi-parametric Imaging using Multiple-echoes with Optimized Simultaneous Acquisition for highly-efficient quantitative MRI”, *Magnetic Resonance in Medicine* (2025)

- C. Oh, W. Jung, **J. Kim**, H. Jeong, and J. Lee, “A fair comparison in deep learning-based QSM reconstruction”, *IEEE Access* (2025)

- **J. Kim**, M. Kim, S. Ji, K. Min, H. Jeong, H.-G. Shin, C. Oh, R. J. Fox, K. E. Sakaie, M. J. Lowe, S.-H. Oh, S. Straub, S.-G. Kim, and J. Lee, “In-vivo high-resolution χ-separation at 7T”, *NeuroImage* (2025)

- M. Kim, S. Ji, **J. Kim**, K. Min, H. Jeong, J. Youn, T. Kim, J. Jang, B. Bilgic, H.-G. Shin, and J. Lee, “χ-sepnet: Deep neural network for magnetic susceptibility source separation”, *Human Brain Mapping* (2025)

- D. Shin, Y. Kim, C. Oh, H. An, J. Park, **J. Kim**, and J. Lee, “Deep Reinforcement Learning-Designed Radiofrequency waveform in MRI,” *Nature Machine Intelligence* (2021)

Under Review

- **J. Kim**, H. Jeong, T. Kim, E. Jeong, J. Jang, Y. Choi, and J. Lee, “Longitudinal QSM: Enhancing Consistency of Multiple Time Point Susceptibility Maps via Simultaneous Reconstruction”

- E. Zhang, Y. Mori, **J. Kim**, H. Lee, J. Lee, and T. Hanakawa, “Paramagnetic and Diamagnetic MRI Markers of Cognitive Decline in Aged Population explored at 7T”

- Y. Chen, Y. Jun, H. G. Shin, S. Li, S. Fujita, X. Yong, **J. Kim**, et al., “MWF-MIMOSA for Efficient Simultaneous Relaxometry and Myelin Water Fraction Mapping”

Conferences

Oral

- **J. Kim**, H. Jeong, T. Kim, Y. Choi, and J. Lee, “Simultaneous reconstruction of longitudinal QSM data (Longitudinal-QSM)”, 2026 ISMRM Annual Meeting & Exhibition

- **J. Kim** and J. Lee, “DeepRF-Grad: Simultaneous Design of RF Pulse and Slice Selective Gradient Using Reinforcement Learning”, 2021 International Congress on MRI & Annual Scientific Meeting of KSMRM

Poster

- **J. Kim**, C. Oh, A.-M. Oros-Peusquens, N. J. Shah, C.-H. Shon, H. Song, and J. Lee, “Visualization of in-vivo brainstem myeloarchitecture via χ-separation at 7T.”, 2025 ISMRM Annual Meeting & Exhibition

- **J. Kim**, H.-G. Shin, M. Kim, S. Ji, K. Min, H. Jeong, S.-G. Kim, and J. Lee, “In-vivo high-resolution χ-separation (chi-separation) at 7T”, 2024 ISMRM Annual Meeting & Exhibition

- **J. Kim** and J. Lee, “Simultaneous Multislice Pulse Optimization Using DeepRF”, 2023 International Congress on MRI & Annual Scientific Meeting of KSMRM

- **J. Kim**, H.-G. Shin, S. Ji, H. Jeong, H. An, C.-H. Sohn, S. Han, H. Song, J.-H. Chae, S.-J. Choi, J.-J. Sung, and J. Lee, “A feasibility study of susceptibility source separation via chi-separation in amyotrophic lateral sclerosis patients at 7T”, 2023 ISMRM Annual Meeting & Exhibition

- **J. Kim**, H. An, C. Oh, B. Bilgic, and J. Lee, “DeepRF for simultaneous multislice pulses”, 2023 ISMRM Annual Meeting & Exhibition

- **J. Kim** and J. Lee, “Accelerated DeepRF using optimal control”, 2022 International Congress on MRI & Annual Scientific Meeting of KSMRM

- **J. Kim**, D. Shin, H. An, H. Jeong, M. Kim, and J. Lee, “Accelerated DeepRF using modified optimal control”, 2022 ISMRM Annual Meeting & Exhibition

- **J. Kim**, D. Shin, J. Park, H. Jeong, and J. Lee, “DeepRF-Grad: Simultaneous design of RF pulse and slice selective gradient using self-learning machine.” 2021 ISMRM Annual Meeting & Exhibition

Teaching Experiences

- Teaching Assistant, Bioelectrical Information Engineering (undergraduate course), Fall semester, 2023, Seoul National University, Korea

- Teaching Assistant, Signals and Systems (undergraduate course), Spring semester, 2021, Seoul National University, Korea

- Teaching Assistant, Topics in Communications (graduate course), Spring semester, 2020, Seoul National University, Korea

Professional Experiences

- ISMRM & ISMRT Annual Meeting & Exhibition, 07-12 May 2022, London, UK.
  - *Moderator*, Oral Session: RF Pulse Design, Parallel Transmission & B1 Shimming

Skills

- Programming: MATLAB, Python, Linux

- MRI Systems: SIEMENS 3T and 7T operation

- MRI Sequence Development: Pulseq-based sequence programming

- NeuroImaging Tools: FSL, ANTs, SPM

- Machine Learning: Pytorch, Tensorflow